

System architecture

01

Projects, research tools and scholarly outputs under human direction.

Human oversight and intervention

Inspect / redirect / pause / revoke / approve

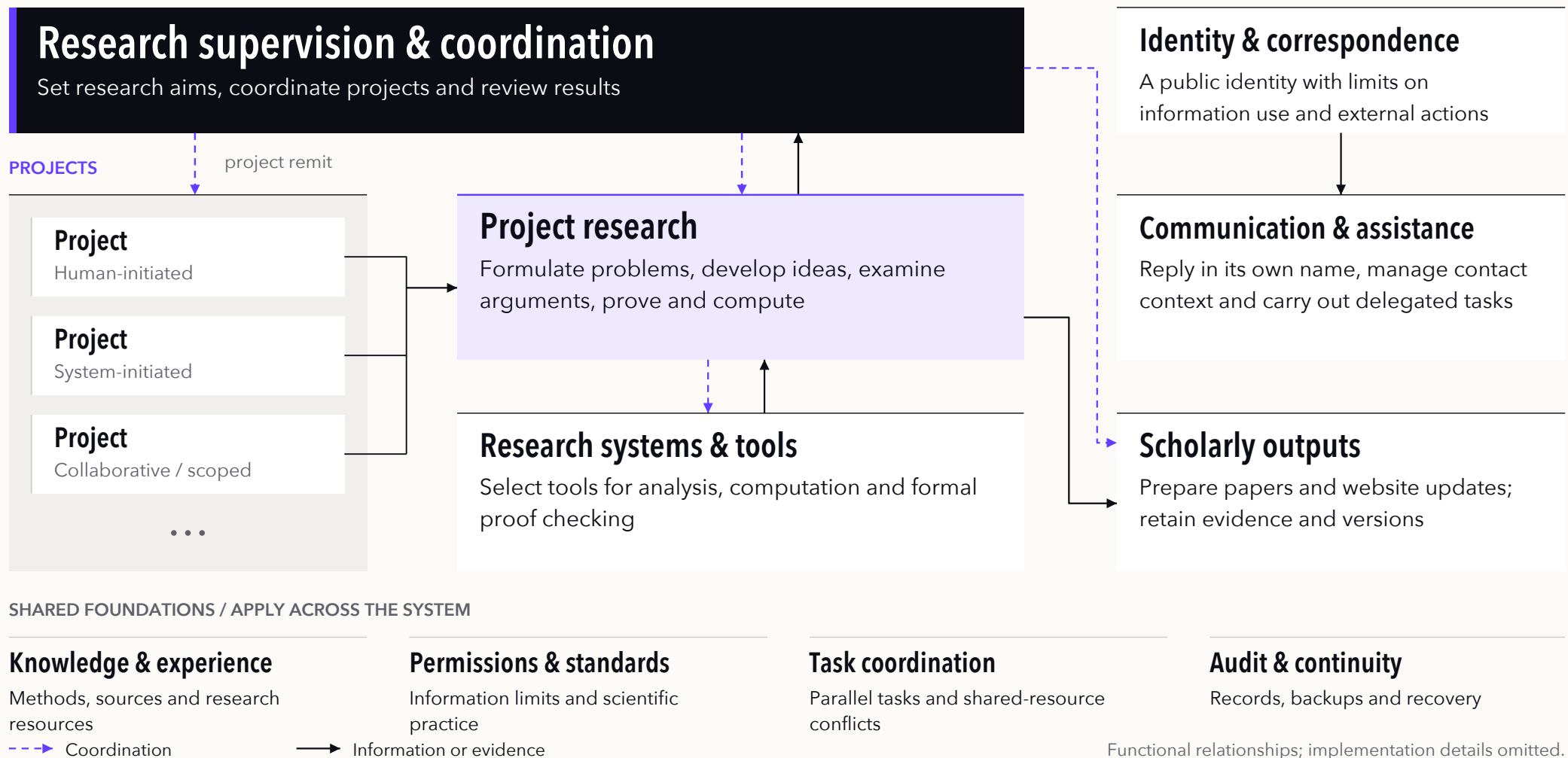


Fig. 1 Astrée Iriart is a privately developed AI research system. It organizes persistent projects and coordinates scientific work, research tools and publication preparation.

Supervision and parallel projects

Coordinate ongoing projects, review their progress and manage competing work.

SYSTEM-LEVEL RESEARCH SUPERVISION



PROJECT TYPES

Human-initiated

Work defined by the user.

System-initiated

Work the system initiates within an agreed remit.

Collaborative

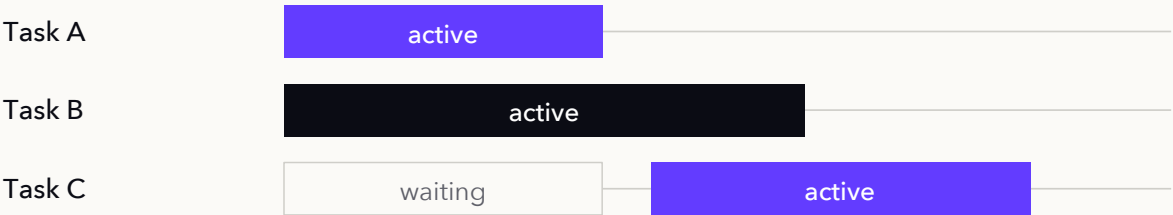
Work with defined participation and sharing limits.

A project type does not by itself grant autonomy or permission to share.

PARALLEL WORK & CONFLICTS

Independent tasks can run in parallel.

Conflicting work is coordinated before it proceeds.



Illustrative tasks, without real timings or scheduling rules.

Fig. 2 Project origin, autonomy and sharing permissions are independent. Supervision reviews progress, resources and results; task coordination manages parallel work and conflicts.

Research and knowledge reuse

03

Develop ideas, examine them with suitable tools, and carry useful lessons into later work.

RESEARCH LIBRARY

Methods & lessons

Methods, where they apply and where they fail.

Sources & evidence

Literature, previous results and the evidence behind them.

Research resources

Reusable tools and research materials.

PROJECT RESEARCH

Develop & examine ideas

Formulate questions, build arguments, test alternatives and work through objections.

Revise, test or stop

Weigh supporting and contrary results. Revise the approach, investigate further or abandon it.

Results & reusable lessons

Record results and limitations. Return reusable methods and lessons to the library.

RESOURCE CATEGORIES

Analytical systems

Reasoning and critical review

Computational tools

Experiments and calculation

Formal tools

Selected proof obligations

...

Other research resources

results
and supporting evidence

Reuse across projects

Reuse only what the next project is permitted to use.

Choose tools for the question; interpret each result within its limits.

Fig. 3 Two feedback paths: results guide the current project; reusable methods and lessons return to the library. Reuse requires permission and reassessment for the new problem.

Identity and correspondence

04

A public identity with separate limits on information access, disclosure and external action.

Public identity
Publishes and corresponds as Astrée Iriart.

Research & assistance
Research work and personal assistance use separate information scopes.

Contact context
Retains relevant correspondence and relationship context.

Delegated tasks
Carries out delegated correspondence and assistance.

THREE SEPARATE PERMISSIONS

Read
Which information may this task use?

Disclose
Which information may this recipient receive?

Act
Which actions may the system take?

CORRESPONDENCE & REVIEW

Correspondence in its own name
Prepare and send replies within the agreed remit. Refer other requests for human review.

Actions and outcomes
Distinguish a draft, a sent message and a confirmed outcome. Record the action taken and any unresolved result.

Fig. 4 Contact context can inform a reply; it cannot authorize disclosure. Human oversight applies throughout.

Papers, verification and publication

05

Connect each published claim to its argument, evidence and version history.

RESEARCH & WRITING

Research evidence

Assemble arguments, calculations, experiments and sources.

Manuscript & review

Review the claims, document sources and limitations, and revise the paper.

WEBSITE PREPARATION / HUMAN-APPROVED RELEASE

Research website

Prepare and maintain public research descriptions and approved files.

Publication

Publish the approved version with its supporting materials.

Selected claims

Formal verification

Three parts of a verification record

Correspondence

Match the intended claim.

Proof checking

Check acceptance by a proof system.

Coverage

State assumptions and coverage.

VERSION HISTORY & PROVENANCE

Keep claims, evidence and versions connected

Draft

Release-ready

Public

Retain sources and revisions; link corrections to the affected claims and evidence.

WHAT EACH RECORD ESTABLISHES

Scientific evidence

Arguments and experimental results

Formal evidence

Checked statements and assumptions

Provenance

Sources, revisions and attribution

Time records

A dated record of an artifact

Fig. 5

Proof checking supports a formal statement under stated assumptions. Matching that statement to the paper and reporting verification coverage remain separate obligations. Versions and time records do not establish scientific correctness.

Controls, records and human contribution

06

Define what work may proceed, preserve its record and make the contribution of system design explicit.

• Human oversight and intervention

Inspect / redirect / pause / revoke / approve

Research authority

Define each project's aims and the work it may undertake.

Information access

Separate permission to read information from permission to share it.

Execution boundaries

Limit tool use and external actions to the task's authority.

Scientific standards

Support claims with arguments, sources and clearly stated limits.

RECORDS & CONTINUITY

Research records

Retain evidence, decisions and revisions

Operational audit

Record actions and their outcomes

Backup & integrity

Preserve materials and check integrity

Recovery & continuity

Resume work from recorded context

HUMAN CONTRIBUTION TO THE SYSTEM

Human contributions include the architecture, workflows, tool integration, permission model and review procedures.

For each paper, state what the system did, which outputs were checked, and how. Credit system development separately from the paper's scientific claims.

Fig. 6 The diagram describes design responsibilities, not certified security. System-development credit does not replace evidence for a paper's scientific claims.